

G9 Scope: what this work establishes, and what it does not

ESTABLISHES

- Coupled MPM flood-vehicle simulation, 17 runs, geometry gates passed.
- is_gaussian_ply guard routes watertight meshes to solidify_watertight, giving 100.00 pct mesh containment on the sampled gate. This is an ORIGINAL PATCH to the upstream dispatch and it is a contribution, not a workaround.
- D x V is bit-identical at 0.441644 m2/s across a 2.1x mass range, because the criterion has no mass term.
- Velocity sweep: D x V returns the wrong verdict at 2 of 6 points on the very axis it is built on. One of the two, v = 1.0 m/s, is robust to any onset threshold up to 0.2249 m. The second, v = 0.5 m/s, clears the chosen 0.05 m by 0.0078 m and is reported as marginal.
- Local bow depth exceeds the AR&R 0.30 m cap in 19 of 20 runs, and the exceedance grows with mass. The exception is v = 0.5 m/s at 0.295361 m.
- Determinism verified across the 17 runs that record it; zero out-of-bounds particle frames across all 20, a count taken after escaped particles are clipped back in.

DOES NOT (scope, not apology)

- No gsplat-reconstructed scene entered a simulation.
- Vehicle velocity is zero in every run. This is a stationary-vehicle study, which is what AR&R itself measures.
- One hull, three masses. NOT three vehicles. Class names denote only which AR&R limit set was applied.
- Realized density 310 / 453 / 658 kg/m3 is ABOVE the project's own stated 100 to 300 kg/m3 band.
- Water particle passthrough 7.3 to 15.9 pct, worst at v = 3.0 m/s.
- Mach 0.039 to 0.234 across the velocity sweep, 0.117 at the 1.5 m/s baseline. The fastest runs sit 2.3x outside the usual weakly-compressible 0.1 limit.
- No Grid Convergence Index is computable: the g64 to g96 change is negative and non-monotonic.
- ~~Track 2 null result: gap 0.295 m at v = 0, no contact, verdicts retracted.~~